NIHAL POTHUNOORI

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EDUCATION

University of Illinois at Urbana-Champaign Expected May 2027
B.S. Electrical Engineering - Urbana-Champaign, IL
Relevant Coursework: Analog Signal Processing; Semiconductor Electronics; Computer Systems & Programming

SKILLS

Programming: Python, C/C++, Matlab, Java, C#, .NET
Electrical: Analog & digital circuits, PCB/PCBA design (schematic + layout), I2C/SPI, op-amp signal conditioning, power regulation, embedded bring-up & debugging
Tools: Altium Designer, KiCad, LTSpice, Git, Arduino, Raspberry Pi, ROS2, OpenCV, TensorFlow/PyTorch; oscilloscope, multimeter, bench power supply
Fabrication: Soldering, rapid prototyping, troubleshooting & failure analysis

WORK EXPERIENCE

Undergraduate Researcher, DASLABs Jan 2026 - Present
University of Illinois Champaign, IL
- Implementing vision-language-action (VLA) models to identify stable rest/anchor locations for a soft robot during locomotion and manipulation tasks.
- Built a ROS2-based multimodal perception + evaluation pipeline in Python (RGB/RGB-D segmentation, logging, metrics) to benchmark rest-point selection reliability.

Undergraduate Researcher, Autonomous Materials Systems Group Aug 2025 - Present
University of Illinois Champaign, IL
- Designed and built a robotic manipulator + gantry system for automated DCPD polymer sample preparation in a DARPA-funded lab project.
- Applied TensorFlow-based reinforcement learning for closed-loop process control; improved throughput by reducing sample prep time 75%.

Web Development / IT Support Aug 2025 - Present
College of Education, UIUC Champaign, IL
- Built a .NET 8 C# library integrating with Azure Active Directory and automated permissions management for ~2,500 users; shipped web updates in HTML/C#.

PROJECTS

Search & Rescue Robot Jun 2025 - Present
Personal Project
- Prototyping an autonomous ground robot to navigate around falling/shifting debris using real-time perception and planning.
- Implemented vision-based obstacle avoidance and A* path planning on Raspberry Pi; structured ROS2 nodes for future LiDAR-SLAM integration.

Laundry Robot Jun 2024 - Jul 2024
Personal Project
- Built an autonomous mobile system to collect scattered clothing using an OpenCV pipeline and closed-loop control on a differential-drive platform.
- Designed trajectory planning + PID wheel control with encoder feedback in embedded C/C++; tuned parameters using logged telemetry.

Wearable EMG Instrument (2nd Place - International IEEE Arduino Contest) Aug 2023
Personal Project
- Designed an EMG analog front-end (op-amps, band-pass filtering, ADC) and mapped muscle activation to musical notes.
- Prototyped/soldered circuitry and debugged noise/stability with oscilloscope and bench supply; iterated filter stages in LTSpice/Matlab.

LEADERSHIP

Hardware Lead, iRobotics Micromouse Jan - May 2025
University of Illinois Champaign, IL
- Led hardware design for maze-solving robots; ran weekly design reviews and taught PCB/PCBA fundamentals (15+ members).
- Fabricated custom PCBs in Altium with I2C/SPI sensor interfaces and analog filtering; performed soldering/rework and bench debugging.